START TECH LIMITED

PYTHON PROGRAMMING

MODULE 1: INTRODUCTION TO PYTHON PROGRAMMING

LEARNING OBJECTIVES

At the end of this module, you should be able to:

• Define Python.
• Explain the history of Python.
• State the features of Python.
• Describe the applications of Python.
• Write and run your first Python program.

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1. WHAT IS PYTHON?

Python is a high-level, interpreted, and general-purpose programming language. It is one of the easiest programming languages to learn because of its simple and readable syntax.

Python is widely used for developing websites, automating repetitive tasks, analyzing data, creating artificial intelligence systems, developing desktop software, and cyber security applications.

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2. HISTORY OF PYTHON

Python was created by Guido van Rossum in the late 1980s and officially released in 1991.

The name "Python" was inspired by the comedy television show "Monty Python's Flying Circus."

Today, Python is one of the most popular programming languages in the world.

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3. FEATURES OF PYTHON

• Easy to learn
• Easy to read
• Open source
• Cross-platform
• Interpreted language
• Object-Oriented Programming (OOP)
• Large standard library
• Portable
• Powerful and flexible

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4. APPLICATIONS OF PYTHON

Python is used in many fields including:

• Web Development
• Data Analysis
• Artificial Intelligence (AI)
• Machine Learning
• Automation
• Cyber Security

• Desktop Applications
• Scientific Computing
• Networking
• Cloud Computing

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5. ADVANTAGES OF PYTHON

• Simple syntax
• Faster software development
• Huge community support
• Thousands of free libraries
• Easy debugging
• Works on Windows, Linux and macOS
• Suitable for beginners and professionals

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6. DISADVANTAGES OF PYTHON

• Slower than compiled languages such as C++
• Uses more memory
• Less suitable for mobile application development

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7. INSTALLING PYTHON

Step 1:
Visit https://www.python.org

Step 2:
Download the latest version of Python.

Step 3:
Run the installer.

Step 4:
Tick the option:

✓ Add Python to PATH

Step 5:
Click Install Now.

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8. PYTHON IDLE

IDLE stands for Integrated Development and Learning Environment.

It allows you to:

• Write Python programs
• Run Python programs
• Save programs
• Edit programs
• Debug programs

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9. YOUR FIRST PYTHON PROGRAM

Example:

```
print("Hello World")
```

Output:

Hello World

Another Example:

```
print("Welcome to START TECH LIMITED")
```

Output:

Welcome to START TECH LIMITED

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10. CAREER OPPORTUNITIES

After learning Python you can become:

• Python Developer
• Web Developer
• Backend Developer
• Data Analyst
• Data Scientist
• Machine Learning Engineer
• AI Engineer
• Automation Engineer
• Cyber Security Analyst
• Software Engineer

MODULE SUMMARY

In this module you have learned:

✓ What Python is
✓ History of Python
✓ Features of Python
✓ Applications of Python
✓ Advantages
✓ Disadvantages
✓ Installing Python
✓ Python IDLE
✓ Your First Program
✓ Career Opportunities

PRACTICAL EXERCISE
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1. Define Python.

2. Who developed Python?

3. In which year was Python officially released?

4. State five features of Python.

5. State five applications of Python.

6. Write a Python program that displays:

Welcome to START TECH LIMITED

7. Explain the purpose of the print() function.

8. List five careers that require Python programming skills.

END OF MODULE 1
